

Boss Challenge — Paper 41 (Class 7)

Motion & Time | Respiration in Organisms | Comparing Quantities | Simple Equations

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. The distance that a moving object covers in each single unit of time is what we call its:
(A) speed
(B) distance
(C) weight
(D) time
2. Inside every living cell, food is broken down to set free the energy stored in it; this life process is called:
(A) digestion
(B) transpiration
(C) germination
(D) respiration
3. A comparison of two quantities of the same kind, made by dividing one by the other, is called a:
(A) product
(B) sum
(C) ratio
(D) difference
4. A statement that two expressions are equal and that contains an unknown to be found is called an:
(A) graph
(B) average
(C) equation
(D) ratio
5. The standard SI unit in which the speed of an object is measured is the:
(A) kilogram
(B) metre per second (m/s)
(C) second
(D) metre
6. Breathing means taking air into the body and giving it out; one breath is made of one inhalation and one:
(A) respiration
(B) digestion
(C) circulation
(D) exhalation

7. The ratio 2 : 5, when written as a fraction, is:

- (A) $\frac{5}{2}$
- (B) $\frac{7}{2}$
- (C) $\frac{2}{7}$
- (D) $\frac{2}{5}$

8. In the equation $x + 5 = 12$, the letter standing for the value we must find is called the:

- (A) constant
- (B) variable (the unknown)
- (C) answer
- (D) equals sign

9. A car's speedometer shows how fast it is going, while the total distance it has covered is read from the:

- (A) thermometer
- (B) barometer
- (C) odometer
- (D) speedometer

10. During breathing, the gas that our body takes in from the air is:

- (A) carbon dioxide
- (B) nitrogen
- (C) hydrogen
- (D) oxygen

11. The phrase 'per cent' means out of every:

- (A) hundred
- (B) ten
- (C) fifty
- (D) thousand

12. Find the value of the unknown in the equation $x + 5 = 12$. It is $x =$:

- (A) 60
- (B) 7
- (C) 17
- (D) 5

13. Motion in which an object covers equal distances in equal intervals of time is called:

- (A) circular motion
- (B) uniform motion
- (C) no motion
- (D) non-uniform motion

14. The gas that the body gives out as a waste product of respiration is:
- (A) water vapour
 - (B) nitrogen
 - (C) oxygen
 - (D) carbon dioxide
15. To change a fraction into a percentage, you multiply the fraction by:
- (A) 10
 - (B) 2
 - (C) 1000
 - (D) 100
16. To solve the equation $x - 3 = 10$, you add 3 to both sides, which gives x equal to:
- (A) 10
 - (B) 7
 - (C) 13
 - (D) 30
17. A car that keeps speeding up, slowing down and changing its pace along a busy road is showing:
- (A) rest
 - (B) steady motion
 - (C) non-uniform motion
 - (D) uniform motion
18. Respiration that uses oxygen to break down glucose and release energy is called:
- (A) anaerobic respiration
 - (B) transpiration
 - (C) aerobic respiration
 - (D) photosynthesis
19. Expressed as a percentage, one quarter - that is the fraction $\frac{1}{4}$ - comes to:
- (A) 75%
 - (B) 14%
 - (C) 25%
 - (D) 4%
20. The equation $2x = 14$ is solved by dividing both sides by 2, which gives x equal to:
- (A) 16
 - (B) 12
 - (C) 7
 - (D) 28

21. To work out the speed of a moving body you divide the distance travelled by the:
- (A) weight of the body
 - (B) number of wheels
 - (C) starting point
 - (D) time taken
22. A kind of respiration that happens in the complete absence of oxygen is known as:
- (A) inhalation
 - (B) anaerobic respiration
 - (C) aerobic respiration
 - (D) digestion
23. Fifty per cent (50%) of any quantity is the same as the fraction:
- (A) $\frac{1}{2}$
 - (B) $\frac{5}{1}$
 - (C) $\frac{1}{5}$
 - (D) $\frac{1}{4}$
24. Solving $x/4 = 5$ by multiplying both sides by 4 gives x equal to:
- (A) 1.25
 - (B) 20
 - (C) 9
 - (D) 5
25. A train travels 180 km in 3 hours. Its average speed for the journey is:
- (A) 540 km/h
 - (B) 30 km/h
 - (C) 183 km/h
 - (D) 60 km/h
26. During very hard exercise our muscles may respire without enough oxygen, building up a substance that causes cramps, namely:
- (A) water
 - (B) oxygen
 - (C) glucose
 - (D) lactic acid
27. About 1 part in every 5 of the air we breathe is oxygen. Written as a percentage, this share of oxygen is:
- (A) 20%
 - (B) 5%
 - (C) 15%
 - (D) 50%

28. Moving a term from one side of an equation to the other while changing its sign is called:

- (A) rounding
- (B) estimation
- (C) multiplication
- (D) transposition

29. A cheetah sprints 100 metres in just 4 seconds. Its speed during the sprint is:

- (A) 4 m/s
- (B) 104 m/s
- (C) 400 m/s
- (D) 25 m/s

30. Yeast respire anaerobically on sugar, producing carbon dioxide and:

- (A) lactic acid
- (B) oxygen
- (C) water
- (D) alcohol

31. When two ratios are equal to each other, the four quantities are said to be in:

- (A) percentage
- (B) difference
- (C) average
- (D) proportion

32. The particular value of the unknown that makes an equation true is called its:

- (A) coefficient
- (B) constant
- (C) solution
- (D) variable

33. One complete to-and-fro swing of a simple pendulum, from one end across to the other and back, counts as one:

- (A) speed
- (B) metre
- (C) oscillation
- (D) second

34. How many full breaths a person takes during one whole minute is measured as the:

- (A) breathing rate
- (B) time period
- (C) pulse
- (D) heart rate

35. If $2 : 3 = 4 : x$, then the value of x is:

- (A) 12
- (B) 7
- (C) 5
- (D) 6

36. A car moving at speed x km/h covers 100 km in 2 hours, giving $2x = 100$. Solving this, x equals:

- (A) 98
- (B) 50
- (C) 102
- (D) 200

37. The time taken by a simple pendulum to complete one full oscillation is known as its:

- (A) frequency
- (B) time period
- (C) distance
- (D) speed

38. An adult human at rest breathes on average about how many times per minute?

- (A) 120 to 150 times
- (B) 15 to 18 times
- (C) 60 to 70 times
- (D) 1 to 2 times

39. A cheetah runs at 30 m/s and a horse at 15 m/s. The ratio of the cheetah's speed to the horse's speed is:

- (A) 3 : 1
- (B) 15 : 30
- (C) 1 : 2
- (D) 2 : 1

40. If a number doubled equals 18, the equation is $2n = 18$, and the number n is:

- (A) 20
- (B) 9
- (C) 36
- (D) 16

41. If a simple pendulum makes 20 complete oscillations in 40 seconds, the time period of one oscillation is:

- (A) 40 seconds
- (B) 2 seconds
- (C) 20 seconds
- (D) 800 seconds

42. The dome-shaped sheet of muscle below the lungs that helps in breathing is the:

- (A) windpipe
- (B) diaphragm
- (C) heart
- (D) rib

43. In a class, 25 students out of 100 chose science as their favourite subject. The percentage who chose science is:

- (A) 100%
- (B) 25%
- (C) 75%
- (D) 2.5%

44. When 7 is added to a number the result is 20. The number is:

- (A) 13
- (B) 140
- (C) 27
- (D) 7

45. On a distance-time graph, an object that stays at rest and does not move at all is shown by a:

- (A) horizontal straight line
- (B) steep rising line
- (C) vertical line
- (D) curved line

46. Air on its way to the lungs passes down the windpipe, which is also called the:

- (A) diaphragm
- (B) artery
- (C) trachea
- (D) oesophagus

47. Thirty per cent (30%) of 200 is:

- (A) 200
- (B) 170
- (C) 60
- (D) 30

48. Three times a number is 21. The number is:

- (A) 63
- (B) 18
- (C) 24
- (D) 7

49. On a distance-time graph, the uniform motion of an object is shown by a:
- (A) zig-zag line
 - (B) straight slanting line
 - (C) circle
 - (D) horizontal line
50. Fish breathe by taking in the oxygen dissolved in water using their:
- (A) skin
 - (B) stomata
 - (C) lungs
 - (D) gills
51. An item bought for Rs 80 and sold for Rs 100 gives a profit of:
- (A) Rs 80
 - (B) Rs 180
 - (C) Rs 100
 - (D) Rs 20
52. A person's breathing rate doubled to 36 after running. If $2r = 36$, the resting rate r was:
- (A) 18
 - (B) 72
 - (C) 34
 - (D) 38
53. Pendulums were once used to run clocks because, for a given pendulum, the time period stays:
- (A) random
 - (B) always decreasing
 - (C) always increasing
 - (D) constant
54. Lacking both lungs and gills, an earthworm takes in oxygen through its damp:
- (A) skin
 - (B) spiracles
 - (C) lungs
 - (D) gills
55. A trader makes a profit only when the selling price of an item is:
- (A) greater than the cost price
 - (B) zero
 - (C) equal to the cost price
 - (D) less than the cost price
56. Solving the equation $3x + 1 = 10$ gives x equal to:
- (A) 30
 - (B) 9
 - (C) 11
 - (D) 3

57. Among these everyday quantities, the one that is a unit of time is the:
- (A) kilogram
 - (B) metre
 - (C) litre
 - (D) second
58. Insects such as grasshoppers take in air through tiny holes along the sides of the body called:
- (A) nostrils
 - (B) stomata
 - (C) spiracles
 - (D) gills
59. A shopkeeper buys a fan for Rs 500 and sells it for Rs 450. This sale results in a:
- (A) no profit and no loss
 - (B) loss of Rs 950
 - (C) profit of Rs 50
 - (D) loss of Rs 50
60. Solving the equation $5x - 2 = 13$ gives x equal to:
- (A) 3
 - (B) 11
 - (C) 65
 - (D) 15
61. A simple pendulum is made of a small heavy ball, the bob, hung from a fixed point by a:
- (A) thread (string)
 - (B) rubber tube full of water
 - (C) metal rod that cannot bend
 - (D) spring
62. In a green plant, gases for respiration pass in and out through tiny pores on the leaves called:
- (A) stomata
 - (B) gills
 - (C) spiracles
 - (D) veins
63. The simple interest on a sum of money depends on the principal, the rate, and the:
- (A) weight of the coins
 - (B) colour of the notes
 - (C) time
 - (D) number of customers
64. If $x/2 + 1 = 5$, then x equals:
- (A) 3
 - (B) 12
 - (C) 8
 - (D) 4

65. Two towns are 240 km apart. A bus that covers the gap in exactly 4 hours travels at an average speed of:
- (A) 40 km/h
 - (B) 244 km/h
 - (C) 60 km/h
 - (D) 960 km/h
66. The carbon dioxide we breathe out can be detected because, when bubbled through it, lime water turns:
- (A) bright red
 - (B) deep blue
 - (C) colourless and clear
 - (D) milky
67. The simple interest on Rs 1000 at 10% per year, for one year, is:
- (A) Rs 1000
 - (B) Rs 10
 - (C) Rs 100
 - (D) Rs 1100
68. To keep an equation true, whatever operation you do to one side you must also do to the:
- (A) other side
 - (B) equals sign
 - (C) answer only
 - (D) first side again
69. An object moving steadily at 10 m/s will, in 5 seconds, cover a distance of:
- (A) 2 metres
 - (B) 50 metres
 - (C) 15 metres
 - (D) 10 metres
70. When we inhale, the diaphragm contracts and moves down, so the space inside the chest cavity:
- (A) stays exactly the same
 - (B) increases
 - (C) decreases
 - (D) fills with blood
71. Air breathed out has about 16% oxygen, while air breathed in has about 21%. The drop in the oxygen percentage is about:
- (A) 21%
 - (B) 16%
 - (C) 5%
 - (D) 37%

72. A pendulum's time period T satisfies the equation $5T = 10$ seconds. Solving it, T equals:

- (A) 5 seconds
- (B) 2 seconds
- (C) 15 seconds
- (D) 50 seconds

73. Since the same time gives a longer distance to a faster body, for a fixed time the distance covered is:

- (A) always equal to the speed
- (B) not related to the speed
- (C) directly proportional to the speed
- (D) inversely proportional to the speed

74. Respiration matters to every living cell chiefly because the process releases:

- (A) energy
- (B) light
- (C) soil
- (D) sound

75. One pendulum has a time period of 2 seconds and another of 1 second. The ratio of their time periods is:

- (A) 2 : 1
- (B) 1 : 2
- (C) 3 : 1
- (D) 2 : 2

76. The sum of a number and 8 is 15. Written as an equation, this is:

- (A) $8x = 15$
- (B) $x + 8 = 15$
- (C) $x - 8 = 15$
- (D) $x + 15 = 8$

77. A sundial tells the time of day by using the slowly moving:

- (A) shadow of a pointer
- (B) flow of sand
- (C) drip of water
- (D) swing of a pendulum

78. The roots of a plant also respire, taking in oxygen from the air that is present in the:

- (A) flowers of the plant
- (B) spaces between soil particles
- (C) deep rock below the soil
- (D) green leaves at the top

79. Convert the decimal 0.6 into a percentage; it becomes:

- (A) 0.6%
- (B) 60%
- (C) 6%
- (D) 600%

80. Twice a number decreased by 3 equals 9, that is $2x - 3 = 9$. Solving it, x equals:

- (A) 6
- (B) 9
- (C) 3
- (D) 12

81. A normal person walking at an easy pace moves at a speed closest to about:

- (A) 0 metres per second
- (B) 100 metres per second
- (C) 25 metres per second
- (D) 1 metre per second

82. When you start to exercise hard, your breathing rate:

- (A) decreases
- (B) increases
- (C) stays exactly the same
- (D) stops completely

83. Out of 50 fish in a tank, 10 died. The percentage of the fish that died is:

- (A) 50%
- (B) 10%
- (C) 20%
- (D) 40%

84. Solving the equation $x + x = 16$ gives x equal to:

- (A) 4
- (B) 8
- (C) 16
- (D) 32

85. A car's odometer reads 5000 km at the start of a trip and 5300 km at the end. The distance travelled on the trip was:

- (A) 300 km
- (B) 5300 km
- (C) 5000 km
- (D) 10300 km

- 86.** A person's breathing rate rose from 18 breaths a minute at rest to 36 after running. It became:
- (A) eighteen times the resting rate
 - (B) the same as the resting rate
 - (C) half the resting rate
 - (D) two times the resting rate
- 87.** The price of an item rises from Rs 100 to Rs 120. The percentage increase in the price is:
- (A) 80%
 - (B) 20%
 - (C) 20
 - (D) 120%
- 88.** If 4 is subtracted from twice a number the result is 10, that is $2x - 4 = 10$. The number x is:
- (A) 14
 - (B) 3
 - (C) 7
 - (D) 6
- 89.** A body that covers 30 m in the first second, another 30 m in the next second and 30 m in the third second is moving with:
- (A) uniform speed
 - (B) increasing speed
 - (C) zero speed
 - (D) decreasing speed
- 90.** A person breathes 18 times each minute. At the same steady rate, the number of breaths in 5 minutes is:
- (A) 90 breaths
 - (B) 18 breaths
 - (C) 13 breaths
 - (D) 23 breaths
- 91.** Before two quantities can be compared as a ratio, they must first be expressed in the:
- (A) decimal form
 - (B) percentage form
 - (C) same unit
 - (D) different units
- 92.** A bus covers d km at 40 km/h for 3 hours, so $d = 40 \times 3$. The distance d is:
- (A) 120 km
 - (B) 40 km
 - (C) 43 km
 - (D) 13 km

- 93.** The hands of an ordinary wall clock move steadily to measure the passing of:
- (A) speed
 - (B) temperature
 - (C) time
 - (D) distance
- 94.** Inside the lungs, the actual exchange of gases happens across millions of tiny balloon-like air sacs called:
- (A) nostrils
 - (B) alveoli
 - (C) stomata
 - (D) spiracles
- 95.** A class has boys and girls in the ratio 3 : 2. The fraction of the class that are boys is:
- (A) $\frac{3}{5}$
 - (B) $\frac{2}{5}$
 - (C) $\frac{3}{2}$
 - (D) $\frac{3}{3}$
- 96.** An equation behaves like a balanced pair of pans on a:
- (A) ruler
 - (B) weighing balance
 - (C) number line
 - (D) clock face
- 97.** Besides metres per second, the speed of vehicles on a road is most commonly stated in:
- (A) kilometres per hour (km/h)
 - (B) kilograms per hour
 - (C) degrees per second
 - (D) litres per minute
- 98.** Compared with aerobic respiration, the anaerobic respiration in our muscles during a sudden burst of activity releases:
- (A) no energy at all
 - (B) exactly the same energy
 - (C) a smaller amount of energy
 - (D) a larger amount of energy
- 99.** Twelve out of every hundred apples in a crate were found to be rotten. The percentage of rotten apples was:
- (A) 12
 - (B) 88%
 - (C) 12%
 - (D) 1.2%

100. The solution of the equation $10 - x = 4$ is x equal to:

- (A) 6
- (B) 14
- (C) 40
- (D) 4